

measurement challenges quickly and accurately, and a digital storage oscilloscope is the best tool to do that.

Why is it important to select the right oscilloscope?

Accurate signal capture is of utmost importance for test and measurement equipment as all your decisions rely on the measured values. You can have correct measurement only when your oscilloscope is compatible with the range of signals you are trying to measure. That is why it is important to have a pretty good idea about the signals that you are going to measure before you buy an oscilloscope. Once you know your signals, you need to understand specifications in the datasheet to select an oscilloscope that will accurately measure all your signals. Also consider its upgradeability to meet your future needs.

Technical specifications

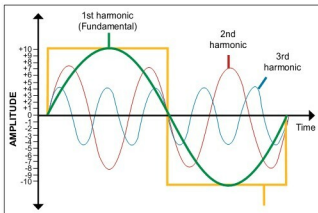
While looking at technical specifications, check whether the values are guaranteed or typical. You cannot completely rely on the typical value of any specification as it will give you only an idea of the oscilloscope's performance. To have the measurement that complies with various quality standards, you need guaranteed specifications.

Technical specifications mentioned in a datasheet are usually difficult to understand. For easy understanding, we can divide specifications into three groups: measurement accuracy, measurement convenience and construction.

Specifications governing measurement accuracy

First, ascertain the signals that you are going to measure. To make sure that the oscilloscope measures all your signals accurately, the below-mentioned specifications of the oscilloscope need to comply with your signals' range.

Bandwidth. Bandwidth indicates an oscilloscope's fundamental ability to measure a signal. It determines the maximum frequency signal that the oscilloscope can accurately measure.



First, second and third harmonics

Accuracy decreases with increase in the signal's frequency. Bandwidth mentioned in the datasheet (say, 100 MHz) is actually the frequency at which a sinusoidal input is attenuated to 70.7 per cent of its true amplitude. Beyond this frequency the oscilloscope cannot support reasonable accuracy.

The oscilloscope must have suitable bandwidth to accurately measure your signals. To decide how much bandwidth you need, find out the range of frequencies you would need to measure, then multiply the maximum frequency by 'five'. For example, to measure a signal with 10MHz frequency accurately, you need an oscilloscope with 50MHz bandwidth. Otherwise, high-frequency changes will not be resolved and the amplitude will be distorted.

In case of digital signals, it is important to capture the fundamental, third and fifth harmonics to display accurate results (as shown in the figure). Therefore the bandwidth of the scope, together with the probe, should be at least five times the frequency of the measured signal.

Rise time. Analogue engineers find bandwidth more interesting, but for a digital engineer rise time is critical. Carefully consider rise time specifications if you will be working with digital signals. Choose an oscilloscope with sufficient rise time to accurately capture the details of fast transitions. An oscilloscope with faster rise time will more accurately capture critical details of a fast transitioning signal.

As a thumb rule, similar to band-

width, the rise time of the oscilloscope should be less than one-fifth of the fastest rise time of the signal to accurately measure it. For example, for a 5ns rise-time signal, an oscilloscope with less than 1ns rise time is required.

Sample rate.

Sample rate refers to how frequently

a digital oscilloscope takes a sample of the signal. The faster the oscilloscope samples, the lesser the details lost while reconstructing the signal. In order to accurately reconstruct a signal and avoid aliasing, the Nyquist theorem states that the signal must be sampled at least twice as fast as its highest-frequency component. This theorem assumes an infinite record length and a continuous signal, but no oscilloscope can offer infinite record length. Therefore sampling at only twice the rate of highest-frequency component will not suffice.

The sampling rate that you require to accurately read your signal will majorly depend on the method used for reconstructing the signal—also called interpolation. For accurate reconstruction using $\sin(x)/x$ interpolation, sample rate should be at least 2.5 times the highest-frequency component of your signal. Using linear interpolation, the sample rate should be at least ten times the highest-frequency signal component.

Record length. The oscilloscope cannot store infinite number of samples as assumed by Nyquist equation. Record length determines the time that can be captured by each channel of the oscilloscope:

Time captured = Record length / Sample rate

Normally, entry-level oscilloscopes come with 2 to 2.5k points, which is more than enough. In general, the greater the record length, the better. To capture and debug serial bus, you need record length as high as 1M points.